

JEE MAIN REVISION MASTER: LIQUID SOLUTION

2026 4 QUESTIONS 6.67% WEIGHTAGE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 32 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Kcet Liquid Solution - Core Application	32 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Liquid Solution - Core Application (Very High Yield — 32 PYQs)

Core Principle: Key problem pattern in Kcet Liquid Solution.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The relative lowering of vapour pressure produced by dissolving 18 g of urea (Molar mass =60g mol⁻¹) in 100 g of water is..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "When 0.0106 mole of acetic acid was dissolved in 1 kg of water, the freezing point depression for this strength of acid was 0.0205 K. ..." [Ans: D]
- [2026 Shift] Q#3 (MCQ): "Match List-I (Laws) with the List-II (Mathematical expressions): List-I List-II (a) Henry's law (i) $p_1 = p_1^0 x_1$ (b) Raoult's law ..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The relative lowering of vapour pressure produced by dissolving 18 g of urea (Molar mass = 60 g mol^{-1}) in 100 g of water is

- (A) 0.025
- (B) 0.512
- (C) 0.051
- (D) 0.250

Correct Answer: None

Solution Analysis: For a non-volatile solute like urea, the relative lowering of vapour pressure is equal to the mole fraction of the solute. $\frac{p^\circ - p}{p^\circ} = x_{\text{(solute)}}$ Moles of urea: $n_{\text{(urea)}} = (18 / 60) = 0.3 \text{ mol}$ Moles of water: $n_{\text{(water)}} = (100 / 18) = 5.56 \text{ mol}$ Now, mole fraction of urea is $x_{\text{(urea)}} = \frac{n_{\text{(urea)}}}{n_{\text{(urea)}} + n_{\text{(water)}} = \frac{0.3}{0.3 + 5.56}$ $x_{\text{(urea)}} = (0.3 / 5.86) = 0.051$ So, the relative lowering of vapour pressure is 0.051 Correct option: Option C.